



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Wednesday, 16 September 2026, 5:43 PM**State** Finished**Completed on** Wednesday, 16 September 2026, 5:44 PM**Time taken** 44 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     for (int i = 0; i < n; i++) {
13         for (int j = i + 1; j < n; j++) {
14             if (a[i] == a[j]) {
15                 printf("%d", a[i]);
16                 return 0;
17             }
18         }
19     }
20
21     return 0;
22 }
23

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2**Started on** Wednesday, 16 September 2026, 5:44 PM**State** Finished**Completed on** Wednesday, 16 September 2026, 5:45 PM**Time taken** 59 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     int slow = a[0];
13     int fast = a[0];
14
15     do {
16         slow = a[slow];
17         fast = a[a[fast]];
18     } while (slow != fast);
19
20     slow = a[0];
21
22     while (slow != fast) {
23         slow = a[slow];
24         fast = a[fast];
25     }
26
27     printf("%d", slow);
28
29     return 0;
30 }
31

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Wednesday, 16 September 2026, 5:46 PM

State Finished

Completed on Wednesday, 16 September 2026, 5:46 PM

Time taken 43 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int T;
5      scanf("%d", &T);
6
7      while (T--) {
8          int n1, n2;
9          scanf("%d", &n1);
10
11          int a[n1];
12          for (int i = 0; i < n1; i++)
13              scanf("%d", &a[i]);
14
15          scanf("%d", &n2);
16
17          int b[n2];
18          for (int i = 0; i < n2; i++)
19              scanf("%d", &b[i]);
20
21          int first = 1;
22

```



```
23 for (int i = 0; i < n1; i++) {
24     for (int j = 0; j < n2; j++) {
25         if (a[i] == b[j]) {
26             if (!first)
27                 printf(" ");
28             printf("%d", a[i]);
29             first = 0;
30             break;
31         }
32     }
33 }
34
35 printf("\n");
36 }
37
38 return 0;
39 }
40 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Wednesday, 16 September 2026, 5:47 PM

State Finished

Completed on Wednesday, 16 September 2026, 5:47 PM

Time taken 38 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int T;
5      scanf("%d", &T);
6
7      while (T--) {
8          int n1, n2;
9          scanf("%d", &n1);
10
11         int a[n1];
12         for (int i = 0; i < n1; i++)
13             scanf("%d", &a[i]);
14
15         scanf("%d", &n2);
16
17         int b[n2];
18         for (int i = 0; i < n2; i++)
19             scanf("%d", &b[i]);
20
21         int i = 0, j = 0, first = 1;
22     }

```

```
23 while (i < n1 && j < n2) {
24     if (a[i] < b[j])
25         i++;
26     else if (a[i] > b[j])
27         j++;
28     else {
29         if (!first)
30             printf(" ");
31         printf("%d", a[i]);
32         first = 0;
33         i++;
34         j++;
35     }
36 }
37
38 printf("\n");
39 }
40
41 return 0;
42 }
43
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2**Started on** Wednesday, 16 September 2026, 5:48 PM**State** Finished**Completed on** Wednesday, 16 September 2026, 5:48 PM**Time taken** 34 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     for (int i = 0; i < n; i++) {
15         for (int j = i + 1; j < n; j++) {
16             if (a[j] - a[i] == k) {
17                 printf("1");
18                 return 0;
19             }
20         }
21     }
22
23     printf("0");
24
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2**Started on** Wednesday, 16 September 2026, 5:49 PM**State** Finished**Completed on** Wednesday, 16 September 2026, 5:49 PM**Time taken** 36 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (i < n && j < n) {
17         if (i == j) {
18             j++;
19         } else if (a[j] - a[i] == k) {
20             printf("1");
21             return 0;
22         } else if (a[j] - a[i] < k) {
23             j++;
24         } else {
25             i++;
26         }
27     }
28
29     printf("0");
30
31     return 0;
32 }
33

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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